

Translated

From: e-maxx.ru

Vertical decomposition

Overview

Vertical decomposition is a powerful technique used in various geometry problems. The general idea is to cut the plane into several vertical stripes with some "good" properties and solve the problem for these stripes independently. We will illustrate the idea on some examples.

Area of the union of triangles

Suppose that there are n triangles on a plane and we are to find the area of their union. The problem would be easy if the triangles didn't intersect, so let's get rid of these intersections by dividing the plane into vertical stripes by drawing vertical lines through all vertices and all points of intersection of sides of different triangles. There may be $O(n^2)$ such lines so we obtained $O(n^2)$ stripes. Now consider some vertical stripe. Each non-vertical segment either crosses it from left to right or doesn't cross at all. Also, no two segments intersect strictly inside the stripe. It means that the part of the union of triangles that lies inside this stripe is composed of disjoint trapezoids with bases lying on the sides of the stripe. This property allows us to compute the area inside each stripe with a following scanline algorithm. Each segment crossing the stripe is either upper or lower, depending on whether the interior of the corresponding triangle is above or below the segment. We can visualize each upper segment as an opening bracket and each lower segment as a closing bracket and decompose the stripe into trapezoids by decomposing the bracket sequence into smaller correct bracket sequences. This algorithm requires $O(n^3 \log n)$ time and $O(n^2)$ memory.

Optimization 1

Firstly we will reduce the runtime to $O(n^2 \log n)$. Instead of generating trapezoids for each stripe let's fix some triangle side (segment $s = (s_0, s_1)$) and find the set of stripes where this segment is a side of some trapezoid. Note that in this case we only have to find the stripes where the balance of brackets below (or above, in case of a lower segment) s is zero. It means that instead of running vertical scanline for each stripe we can run a horizontal scanline for all parts of other segments which affect the balance of brackets with respect to s . For simplicity we will show how to do this for an upper segment, the algorithm for lower segments